

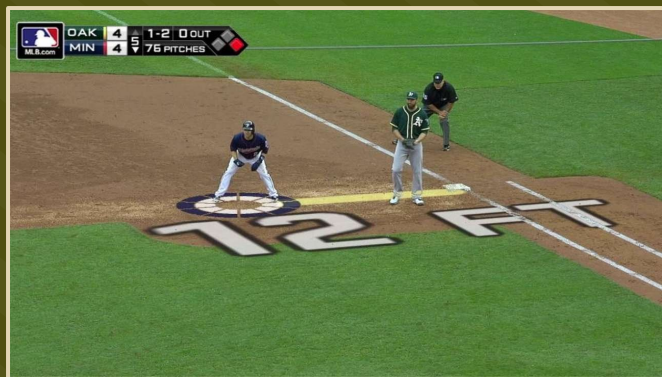


Optimizing Lead Distance

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1. Intro

Lead Distance



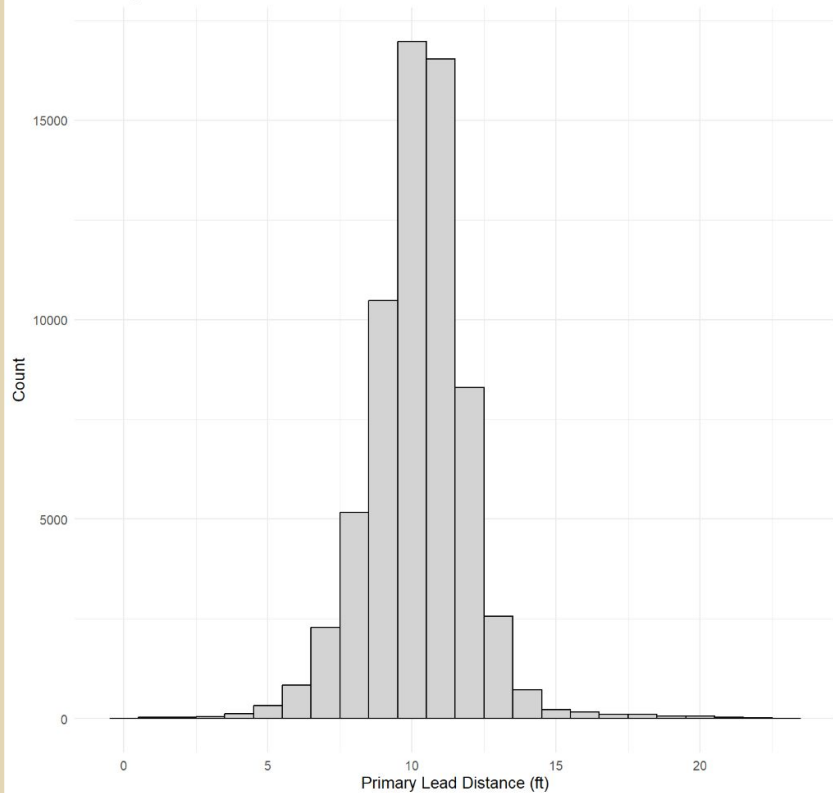
Caught Stealing



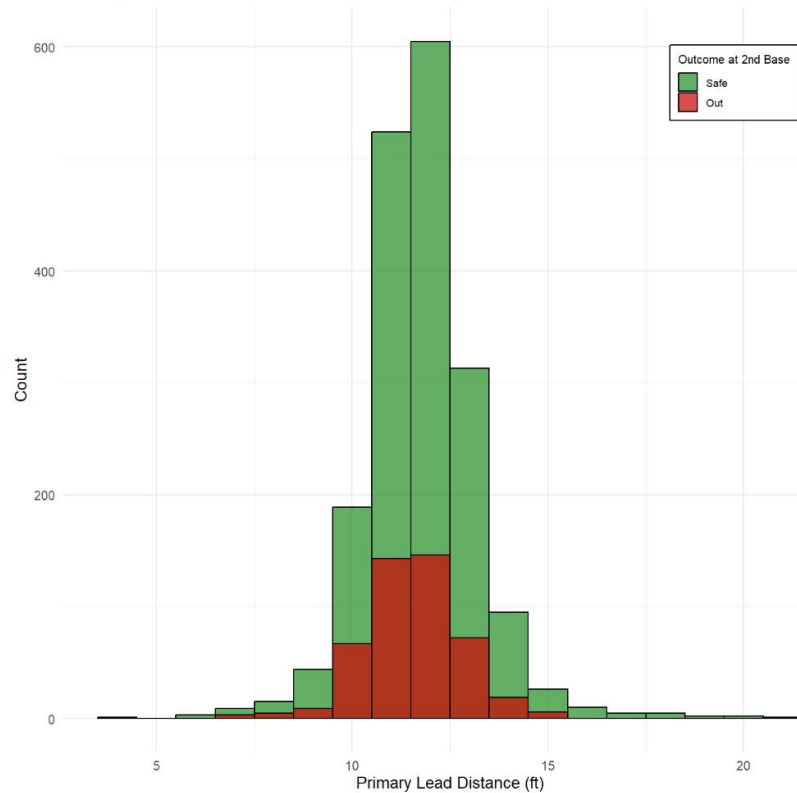
Pickoff



Primary Leads from 1B



Primary Leads from 1B on Stolen Base Attempts



Research Question:

If a runner on 1B intends to steal 2nd,
what is the optimal* lead from first
base?

Outcomes



+0.2 xRuns



-0.45 xRuns

The Formula: $xRuns = 0.2 \times SB - 0.45 \times CS - 0.45 \times PK$

Linear Weights Per Baseball Savant

Factors

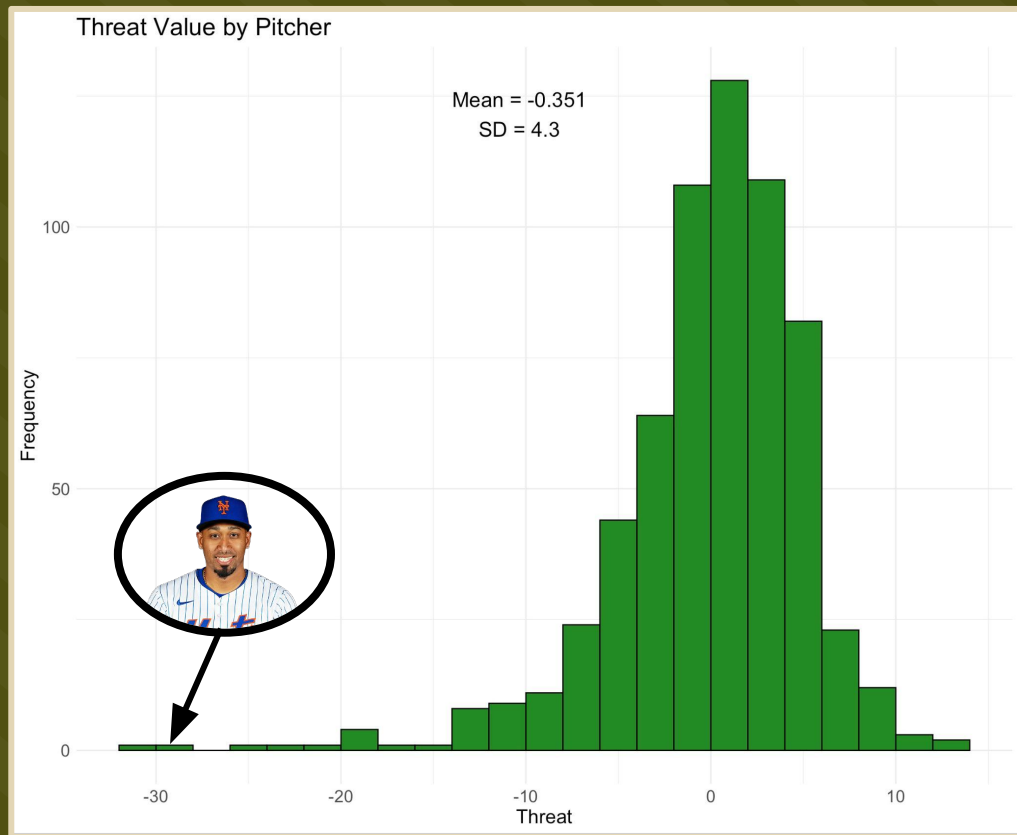
Runner: sprint speed

Pitcher: Threat

Catcher: pop time

Pitcher Threat

Net Bases Prevented
100 Innings Pitched



2. Approach

- Lead Distance
- Threat
- Catcher Pop Time
- Runner Speed



Probabilities

Stolen Base
Caught
Pickoff



xRuns

For one pitch, only lead distance can be controlled...
hold everything else constant, find optimal

Each situation: Own optimal value!

Example Pitch and Model run

	3		▲ 5
P	8	0-1	0 OUT
SKENES		:07	
9. BETHANCOURT 0-1			

Runner: Pete Crow-Armstrong

30 ft/s sprint speed

Pitcher: Paul Skenes

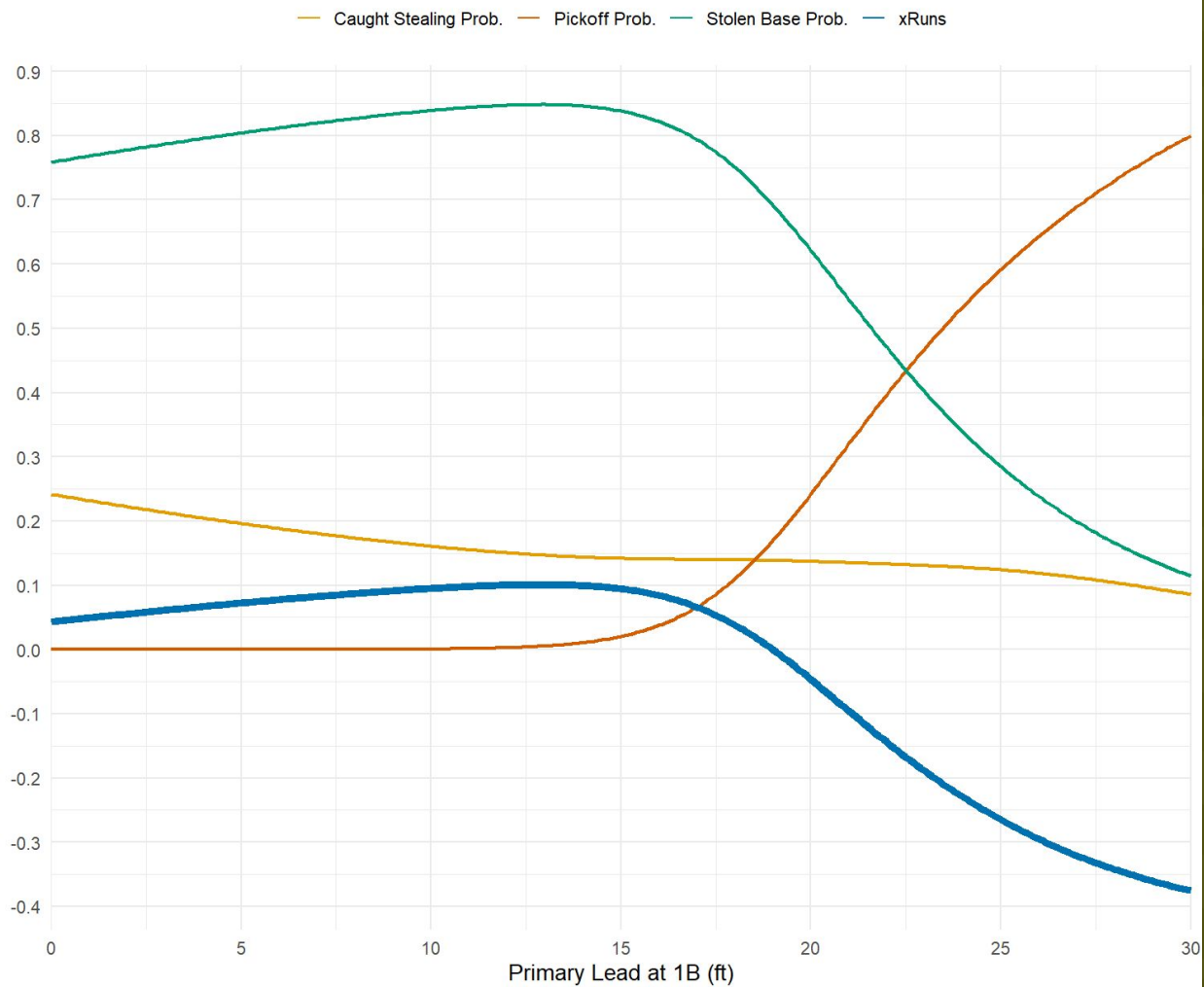
4.03 threat

Catcher: Yasmani Grandal

2.09 s pop time



Model Results: PCA, Skenes, Grandal





Lead

11.5 ft

10.6 ft

Optimal

3. PASQUANTINO 0-2

ADAM P:6

SEASON: .230 AVG, 5 HR, 31 RBI



3 DASQUANTINO 2-2

ADAM D-6



Lead

11.2 ft

14.5 ft

Optimal

BAZ P:27

8. HILL .241

THIS SEASON



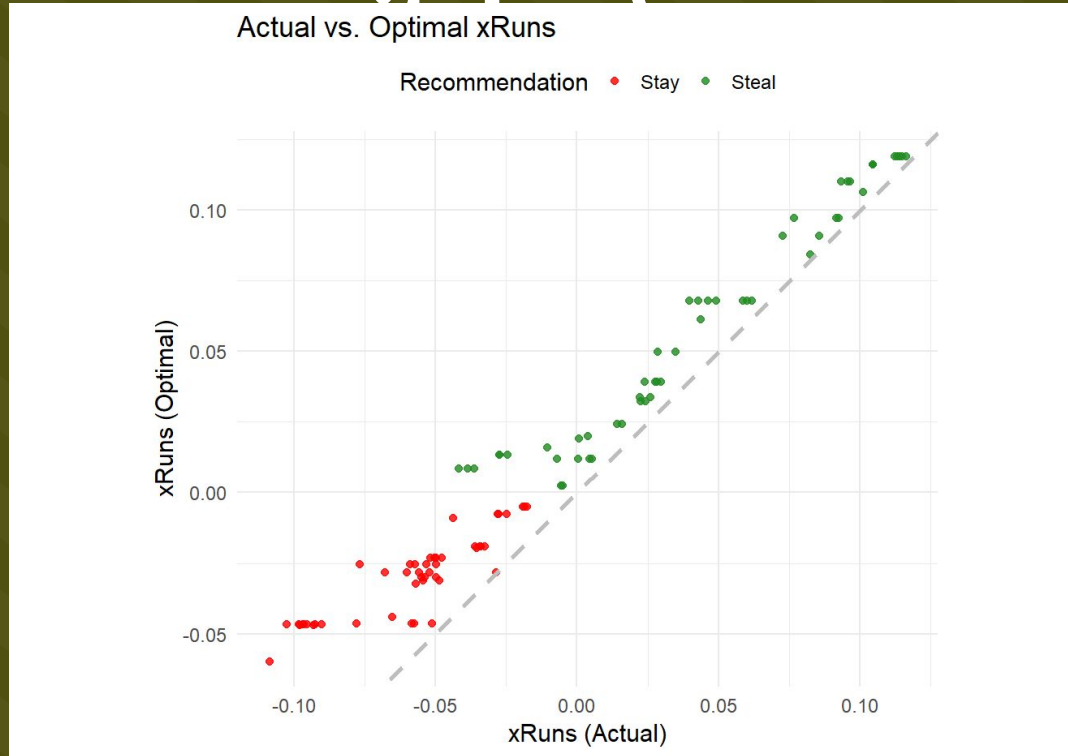
BAZ P:27

8. HILL .241

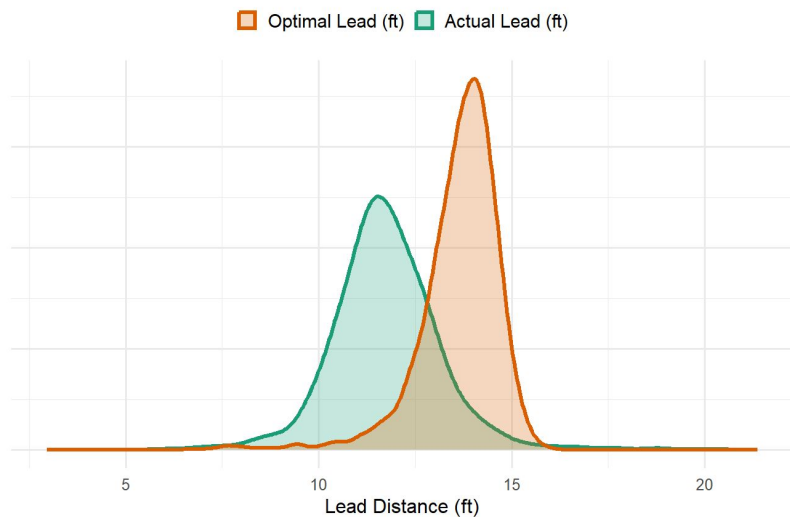
THIS SEASON

3. Results

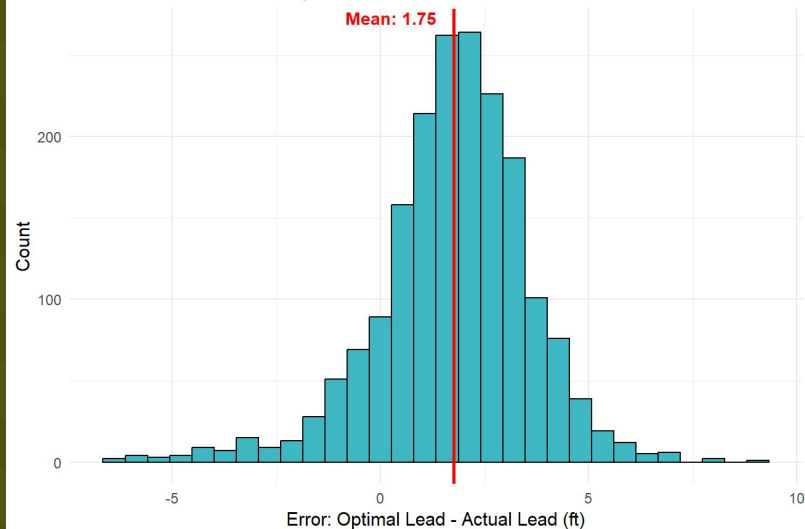
Expected Runs (Sample of 100)



Frequency of leads: Stolen base attempts only



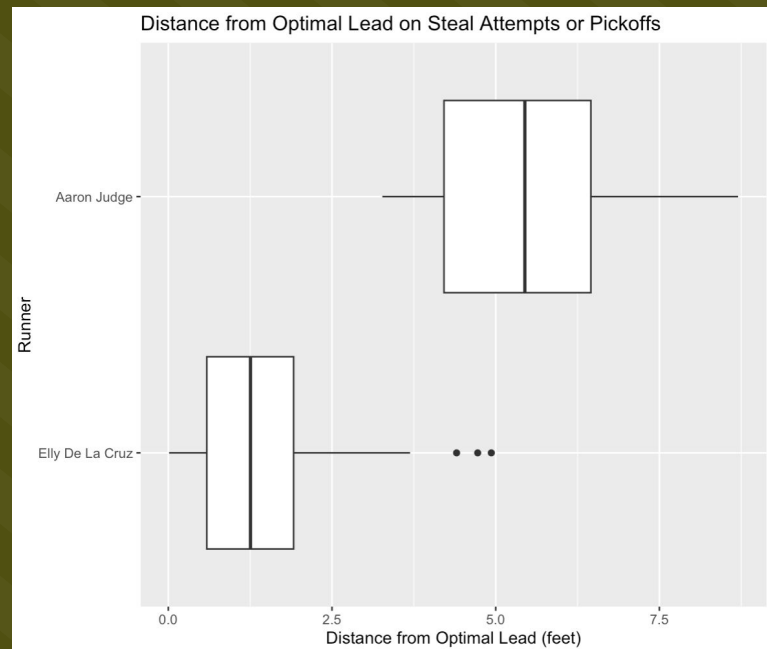
Lead Error on SB attempts



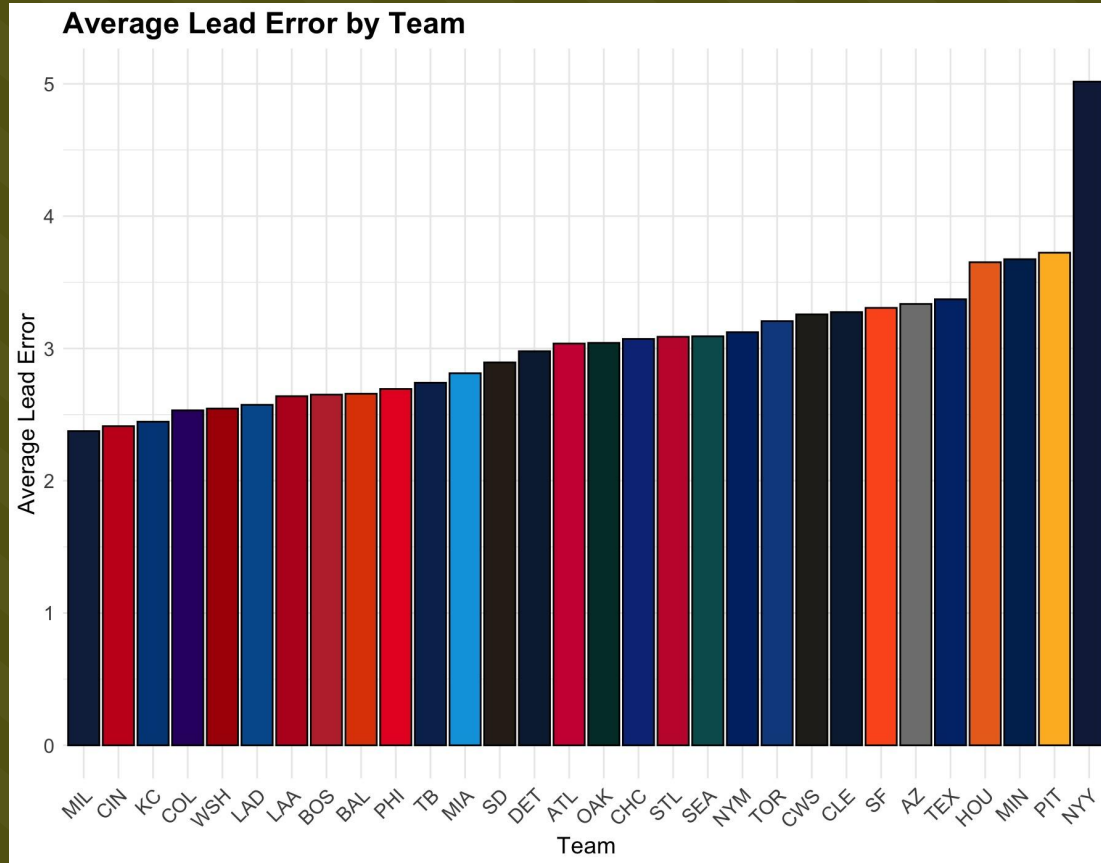
Evaluating Players

	Runner1B	meanChange
1	Forrest Wall	0.8719687
2	Bubba Thompson	1.0144167
3	Jon Berti	1.2039286
4	Joey Wiemer	1.2870000
5	Jacob Young	1.3810897
6	Victor Scott II	1.4370909
7	J.P. Martínez	1.5164286
8	Tyler Wade	1.5521944
9	Tyler Fitzgerald	1.5882741
10	Jorge Mateo	1.6054058
11	Blake Perkins	1.6174976
12	Cole Tucker	1.6525172
13	Elly De La Cruz	1.6551000
14	José Fermín	1.6868533
15	Pete Crow-Armstrong	1.8774651
16	Grant McCray	1.9035238
17	Romy Gonzalez	1.9146637
18	Eli White	1.9325185
19	Weston Wilson	1.9601923
20	Masyn Winn	1.9619541

	Runner1B	meanChange
426	Francisco Alvarez	7.331309
425	Juan Soto	7.322730
424	Nolan Arenado	7.274683
423	Aaron Judge	7.036826
422	Gleyber Torres	6.918720
421	Victor Caratini	6.782630
420	Trent Grisham	6.657861
419	Rowdy Tellez	6.578812
418	Anthony Santander	6.546801
417	Adley Rutschman	6.534206
416	Christian Vázquez	6.288453
415	Jeff McNeil	6.266428
414	Patrick Bailey	6.264222
413	Rafael Devers	6.212691
412	Reese McGuire	6.137299
411	Anthony Volpe	6.092604
410	Pete Alonso	6.064807
409	Ketel Marte	6.040878
408	James McCann	5.940053
407	Yordan Alvarez	5.878298



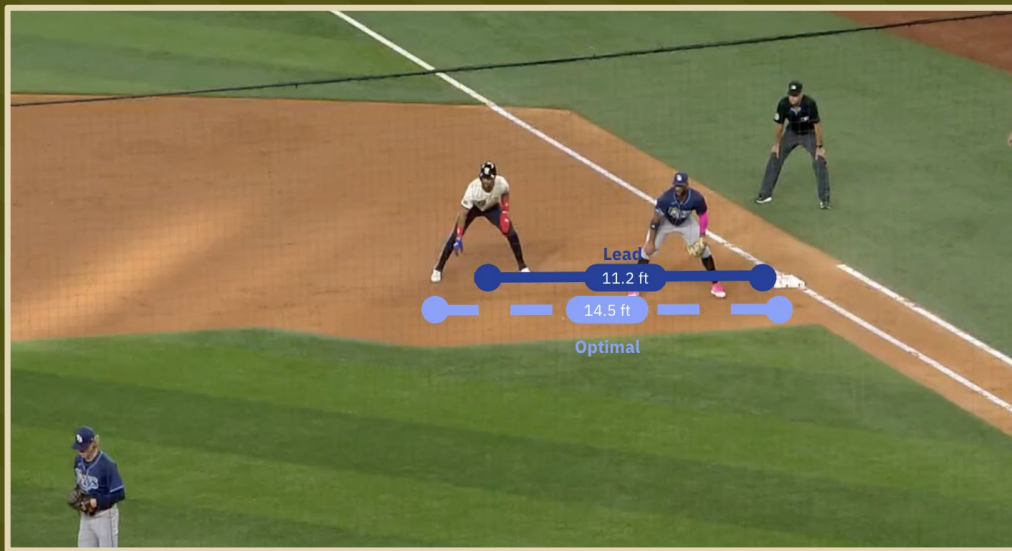
Evaluating Teams



Takeaways

Optimizing lead distance: Just 1 more shuffle → +0.02 runs

Leads are typically too short



Confounding Factors?

Players' fear of making an out on basepath

Game Situation
(Situational vs linear weights)

New rule: 3 pickoffs

Questions

References

https://baseballsavant.mlb.com/leaderboard/basestealing-run-value?game_type=Regular&n=q&pitch_hand=all&runner_moved=All&target_base=All&prior_pk=All&season_end=2025&season_start=2025&sortColumn=simple_stolen_on_running_act&sortDirection=desc&split=no&team=&type=Bat&with_team_only=1&expanded=0

https://baseballsavant.mlb.com/leaderboard/pitcher-running-game?game_type=Regular&n=q&pitch_hand=all&runner_moved=All&target_base=All&prior_pk=All&season_end=2024&season_start=2025&sortColumn=simple_prevented_on_running_attr&sortDirection=desc&split=no&team=&type=Pit&with_team_only=1&expanded=0

https://support.mlb.com/s/?_gl=1*1wh6jhg*_gcl_au*MjAzOTIzNzk0MCAxNzUyNzE4OTA5Ljk3MjI2NzE0My4xNzUzMjI0MzQ5LjE3NTMyMjQzNTQ.*_ga*MTYyOTE1Njc5OC4xNzUyNzE4OTA5*_ga_N8YFCZLYSZ*czE3NTMzMjUwMzgkbzEwJGcwJHQxNzUzMzI1MDM4JGo2MCRsMCRoMjAyMDc2MTE4Mw..

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https://support.mlb.com/s/?_gl=1*1wh6jhg*_gcl_au*MjAzOTIzNzk0MCAxNzUyNzE4OTA5Ljk3MjI2NzE0My4xNzUzMjI0MzQ5LjE3NTMyMjQzNTQ.*_ga*MTYyOTE1Njc5OC4xNzUyNzE4OTA5*_ga_N8YFCZLYSZ*czE3NTMzMjUwMzgkbzEwJGcwJHQxNzUzMzI1MDM4JGo2MCRsMCRoMjAyMDc2MTE4Mw..

Refer

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https://support.mlb.com/s/?_gl=1*1wh6jhg*_gcl_au*MjAzOTIzNzk0MC4xNzUyNzE4OTA5Ljk3MjI2NzE0My4xNzUzMjI0MzQ5LjE3NTMyMjQzNTQ.*_ga*MTYyOTE1Njc5OC4xNzUyNzE4OTA5*_ga_N8YFCZLYSZ*_czE3NTMzMjUwMzgkbzEwJGcwJHQxNzUzMzI1MDM4JG02MCRsMCRoMjAyMDc2MTE4Mw..